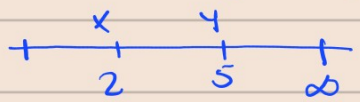
The distance of the matrix

$d(x,y) = |x-y|$



$d|2-5| = 3$

$$\text{distance} = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2}$$

Let X be a non empty set $d: X \times X \to \mathbb{R}$ is a matrix of d satisfies following properties. $\forall x,y,z \in X$

a) $d(x,y) \geq 0$

b) If $d(x,y) = 0 \Leftrightarrow x = y$

c) $d(x,y) = d(y,x)$

d) $d(x,z) \leq d(x,y) + d(y,z)$

If all these properties $\langle X, d \rangle$ is a matrix space.

$d(x,y) = |x-y|$ is $\langle \mathbb{R}, d \rangle$ a matrix space?

i) $|x-y|$ always greater than equal to zero.

ii) $d|x-y| = 0 \Leftrightarrow x, y$

iii) If it is absolute no matter $d(x,y) = d(y,x)$

iv) Absolute Property

Q: $X = R$

learn the proof of all 4 properties

① $d_1(x,y) = (x-y)$ $\langle R, d_1 \rangle$ - Not a matrix space.

② $d_2(x,y) = |x^2 - y^2|$ $\langle R, d_2 \rangle$ - For eg: $x = -1$ and $y = 1$

③ $d_3(x,y) = |x-2y|$ $\langle R, d_3 \rangle$ NO: (i) & (iii)

④ $d_4(x,y) = (x-y)^2$ $\langle R, d_4 \rangle$: NO: (iv)

(1,2,3,4) are not matrix space.

$d(x,y) = \min\{|x|, |y|\}$ - Homework one

$$d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases} \qquad X = \mathbb{R}$$

Is it a matrix space? → yes.

$$d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases} \qquad X = \{a,b,c\}$$

$\langle X, d \rangle$ is a matrix space or not?

$d(a,a) = 0$ $\quad d(b,b) = 0$

$d(a,b) = 1$ $\quad d(c,c) = 0$ - Yes,

$d(b,c) = 1$

Q. $d(x,y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}$ $\quad X = \mathbb{R}$ — Prove

Q):

Let (X, D) is a matrix space, The open ball and closed ball centered at x with a radius of r: $d(x,y) = |x-y|$

$B(x,r) = \{y \in X : d(x,y) < r\}$ - Open Ball

$\bar{B}(x,r) = \{y \in X : d(x,y) \leq r\}$ - closed Ball

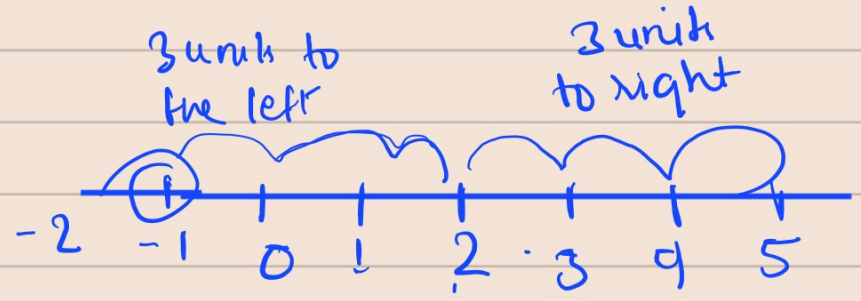
$B(2,3) = ?$

$d(x,4) = |x-4|$

$\Rightarrow |2-4| < 3$

$-1 < 4 < 5$

$(-1, 5)$

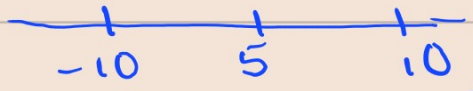


Open Set & Closed Set

Open set :- Let $A \subseteq X$ is said to be open

A is open $\iff \forall x \in A \ \exists r$, s.t $B(x,r) \subseteq A$

$-10, 10)$



$B(5,7)$: ↓ $(-2, 12)$

$B(5,2)$ $(3,7)$ ↳ subset of $(-10,1$

A is open then A^c is closed

$A = (-10, 10)$ Not a subset of $(-10,10)$

$A^c = (-\infty, -10] \cup [10, \infty)$

A is open $\iff \forall x \in A \ r$ s.t $B(x,r) \subseteq A$

Q: lemma if it is true for one, two r then it is true for infinity r.

(1) A matrix space $\langle X, d \rangle$ is defined as follows:-

$X = \{a, b, c\}$

$d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$

$\{a\}$ $\{a,b\}$ $\{a,b,c\}$ Is this open set?

(2) A matrix space $\langle x, d \rangle$ is defined as follows:-

$X = \{a,b,c\}$ $d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$

Find $B(x,r)$?

the open ball of radius r centered at x is

$$B(x,r) = \{y \in X : d(x,y) < r\}$$

1st case: $r \leq 0$

Since distances are not negative, $d(x,y) < r$ can't happen.

$\therefore B(x,r) = \emptyset$

2nd case: $0 < r \leq 1$

for the center x, $d(x,x) = 0 < r$

So x is included

For every point where $y \neq x$

$d(x,y) = 1$

And as $r \leq 1$, $1 < r$ is false.

Therefore, $B(x,r) = \{x\}$

3rd case :- $r > 1$

Now, $d(x, x) = 0 < r$

and for every $y \neq x$

$d(x, y) = 1 < r$

All 3 points are included $B(x, r) = \{a, b, c\}$

$\therefore$

$\therefore B(x, r) = \begin{cases} \emptyset, & r \leq 0 \\ \{x\}, & 0 < r \leq 1 \\ \{a, b, c\}, & r > 1 \end{cases}$

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